3 Partial Derivatives

Definition 3.7.1 If f is a function of two variables, we say that f has a global (or absolute) minimum at (a, b) if $f(x, y) \geq f(a, b)$ for all (x, y) in the domain of f. Similarly, f has a global maximum at (a, b) if $f(x, y) \leq f(a, b)$ for all (x, y) in the domain of f.

Theorem 3.7.2 *(Extreme Value Theorem) If f is a continuous function of two variables, defined on a closed and bounded set D, then f attains an absolute maximum and minimum value on D.*

To find the maximum and minimum values of a continuous function f on a closed and bounded set D, we do the following

1. Find the values of f at the critical points of f in D.
2. Find the extreme values of f on the boundary of D.
3. Identify the largest and smallest of these values.

Example 3.7.3 Find the absulute maximum and minimum values of the function $f(x, y) = x^2 - 2xy + 2y$ on the rectangle $D = \{(x, y) | 0 \leq x \leq 3, 0 \leq y \leq 2\}$.

3.8 Lagrange Multipliers

Example 3.8.1 Find the extreme values of the function $f(x, y) = x^2 + 2y^2$ subject to $x^2 + y^2 = 1$.

Example 3.8.2 A rectangular box without a lid is to be made with 12 m^2 of cardboard. Find the maximum volume of such a box.

Example 3.8.3 Find the points on the sphere $x^2 + y^2 + z^2 = 4$ which are closest to the point $(3, 1, -1)$.

Example 3.8.4 Find the extreme values of the function $f(x, y) = x^2 - 2y^2$ on the disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

Example 3.8.5 Find the extreme values of the function $f(x, y) = e^{-xy}$ subject to the constraint $x^2 + 4y^2 \leq 1$